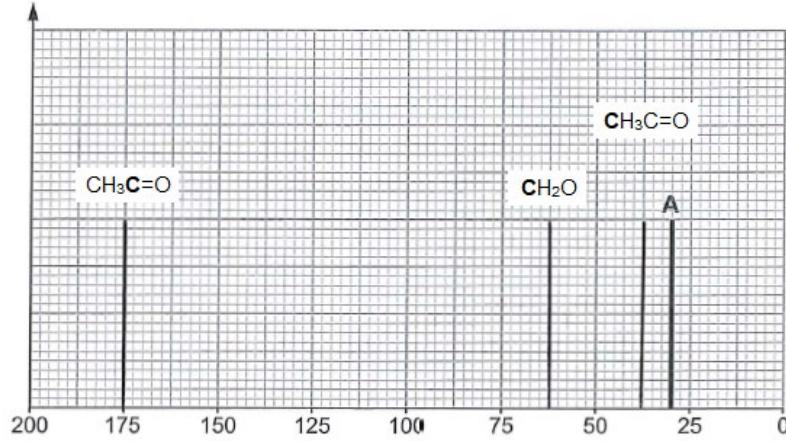


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)			Empirical formula percentage oxygen = 53.33 (1) C : H : O $\frac{40.0}{12} : \frac{6.67}{1.01} : \frac{53.33}{16} \Rightarrow 1 : 2 : 1 \Rightarrow \text{CH}_2\text{O}$ (1) Mass spectrum $M_r = 90$ (1) molecular formula $\text{C}_3\text{H}_6\text{O}_3$ (1) m/z of a fragment linked to identity of fragment (1) e.g. CH_3^+ at 15, COOH^+ / $\text{CH}_3\text{CH}(\text{OH})^+$ at 45, OH^+ at 17 IR spectrum peak in range 1650 to 1750 due to C=O (1) peak in range 2500 to 3200 / 3200 to 3550 due to O—H (1) ^1HNMR 4 hydrogen environments / ratio of 1 : 1 : 1 : 3 (1) any δ value linked to group (1) e.g. 1 to 1.5 due to CH_3, 2.5 to 3.0 due to CH_3CH, 4.0 due to RCOH, 12.5 due to C=OOH Identification X is $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$ / 2-hydroxypropanoic acid do not accept 3-hydroxypropanoic acid		1				
						1				
						1				
							1			
						1				
							1			
							1			
								10		

Question				Marking details	Marks available							
					AO1	AO2	AO3	Total	Maths	Prac		
	(b)	(i)		ignore peak heights accept any sensible approach to labelling peaks award (1) for peak labelled $\text{CH}_3\text{C}=\text{O}$ at 160 to 185 award (1) for peak labelled CH_2O at 50 to 90 award (1) for peak labelled $\text{CH}_3\text{C}=\text{O}$ at 20 to 50 e.g. 			1	1	1	3		
		(ii)		nothing can be deduced from peak size of ^{13}C	1			1				
				Question 11 total	1	6	7	14	0	0		